

Network+ PBQ Cheat Sheet — STP Troubleshooting (VLAN 10/20)

Goal: restore workstation → DB connectivity by fixing STP for VLAN 10 and VLAN 20 on BOTH switches.

Correct solution (what to configure)

- Initialize/enable STP instances for VLAN 10 and VLAN 20 on Switch A and Switch B.
- Success criteria: both switches show VLAN 10 and VLAN 20 in spanning-tree output; ping begins replying.

Commands to run (exact)

```
SwitchA# show vlan
SwitchA# show spanning-tree
SwitchA# spanning-tree vlan 10
SwitchA# spanning-tree vlan 20
SwitchA# show spanning-tree
```

```
SwitchB# show vlan
SwitchB# show spanning-tree
SwitchB# spanning-tree vlan 10
SwitchB# spanning-tree vlan 20
SwitchB# show spanning-tree
```

```
Workstation-1> ping 10.20.20.5 -t
```

How to verify (what you should see)

Check	Where	Expected output cue
VLAN database includes VLANs	Switch A & Switch B	`show vlan` lists VLAN 10 (Workstations) and VLAN 20 (Servers) once configured.
STP instances exist for VLANs	Switch A & Switch B	`show spanning-tree` includes `VLAN0010 STP active` and `VLAN0020 STP active`.
Connectivity restored	Workstation-1 terminal	Ping to `10.20.20.5` changes from `Request timed out.` to `Reply from 10.20.20.5`.

Common PBQ pitfalls

- Configuring VLAN 10/20 on only one switch: BOTH switches must initialize STP for BOTH VLANs.
- Forgetting to validate with `show spanning-tree` after the changes.
- Expecting ping to succeed before both switches have VLAN 10 and VLAN 20 STP active.